WIRELESS INTRUSION DETECTION SYSTEM

Machine learning detection of Wi-Fi deauthentication floods

Student: [YOUR NAME] Course: [YOUR COURSE]

DRAFT / SYNTHETIC DATA ONLY - results are NOT a real Wi-Fi or AWID3 evaluation

OBJECTIVE & SCOPE

Identify normal vs deauthentication-flood activity
in labelled 802.11-like frame records.
Offline CSV replay; no wireless transmission or blocking.
Detection alerts logged with timestamp and source MAC.

DATASET & METHODOLOGY

35,996 generated synthetic frame records; 90 sessions.
Grouped into 1-second windows per source MAC.
75/25 session-level train/test separation.
Random Forest: 100 trees, maximum depth 7.

FEATURES EXTRACTED

1 Number of frames per second
2 Number of deauthentication frames per second
3 Deauthentication fraction
4 Mean received signal strength (RSSI)
5 Management-frame fraction

SYSTEM WORKFLOW

CSV frame records (timestamp, subtype, RSSI, MAC)
|
Per-device, 1-second window aggregation
|
Random Forest classification -> NORMAL / ATTACK
| ATTACK -> console alert + alerts.csv

EVALUATION (SYNTHETIC HOLDOUT)

Training windows: 804 Test windows: 276
Accuracy: 100.0% (276 / 276 correct)
True normal: 156 False alarms: 0
Missed attacks: 0 True attack: 120
These deliberately separable synthetic cases are
NOT evidence of expected real-world accuracy.

CONFUSION MATRIX

LIMITATIONS & IMPROVEMENTS

No live Windows monitor-mode packet capture.
Synthetic traffic is unrealistic and unusually easy.
Next: validate on AWID3 or approved capture; test
noisy RF, spoofed MACs, and overlapping attack traffic.

DELIVERABLES & RESPONSIBLE USE

Code: wids.py, requirements.txt, run_windows.bat
Outputs: metrics.json, confusion_matrix.png, alerts.csv
Traffic is generated locally; no attacks transmitted.
Test live networks only with explicit authorisation.

AI assistance disclosure: ChatGPT helped draft code, test data, evaluation, and poster layout; verify/edit for your own use.

Poster size: A1 landscape (841 x 594 mm) | Replace bracketed details and synthetic results before final submission.